

Surprises in a Classic Boundary-Layer Problem.

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1 Introduction

There are many interesting ODEs in which we can glean many different mathematical techniques. One of the most interesting that occurs is through the evaluation of boundary value problems, most notably, boundary value problems that possess multiple layers on their exterior and interior points. Interestingly enough, there was an ODE proposed by Holmes and later solved by Clark, Gomes, Gonzalez, Stein, and Strogatz that shows just how interesting these layer problems can be. The problem that was initially proposed was the following:

$$\epsilon y'' = yy' - y \tag{1}$$

subject to:

$$y(0) = 1, \quad y(1) = -1$$

2 Holmes: Wrong or right?

One of the most important aspects to understanding the problem is to examine the analysis Holmes in his initial evaluation of the problem. While now we know that there are three distinct solutions, Holmes at the time proposed a single unique solution at $x_0 = \frac{1}{2}$. This is important because it motivates the entire purpose of the problem. This problem, as Holmes points out in his book and alludes to in his partial solution, includes solutions that exist both on the exterior boundary layers and the interior boundary layers. This singular perturbed problem was the first of its kind, or any boundary layer problem for that matter, that had solutions that existed in both these layers.

3 The importance of phase portraits

Here we want to briefly touch on the importance of phase portraits to solve this problem. Essentially, we are boiling the problem into a system of first order

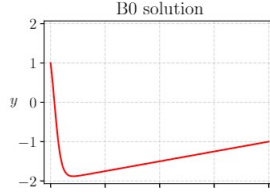


Figure 1: Boundary: $x_0 = 0$

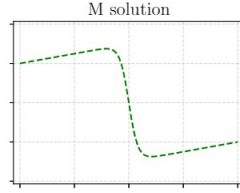


Figure 2: Boundary: $x_0 = \frac{1}{2}$

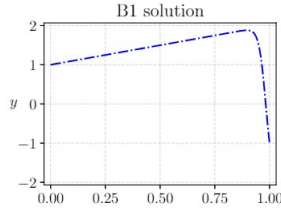


Figure 3: $x_0 = 1$

differential equations that divide the differential equation into trajectories of a vector field:

$$\begin{aligned} y' &= z, \\ z' &= \frac{1}{\epsilon}y(z-1) \end{aligned}$$

If we look at this as a dynamic system, the vector (y', z') tells us about a particle moving with instantaneous velocity at time t for (y, z) . This can be seen graphically below: Clearly we can look at the above vectors as trajectories which there are three types: parabolic trajectories at the line $z = 1$ above the horizontal line, straight trajectories at $z = 1$ on the invariant line, and finally periodic trajectories on $z = 1$ that form closed loops. These trajectories form a bilaterally symmetric loop. This can be proven by integrating the equation $2\epsilon[z + \log(1 - z)] - y_2 = C$ which will be revisited later.

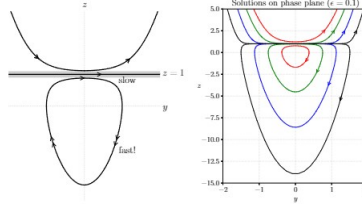


Figure 4: Phase Portraits

4 Perturbation

4.1 The idea of the method of Perturbation Theory

The method we used here is the method of matched asymptotic expansions. We approximate the solution using a series expansion that takes into account the dominant behavior in different regions, the boundary layers. We will get several two different approximate solutions, the outer solution and the inner solution, and then use them to get the composite solution which is valid on the whole range.

4.2 Outer Solution

To get the outer solution, we need to assume the solution can be expanded in the form of powers of ϵ :

$$y(x) \sim y_0(x) + \epsilon y_1(x) + \epsilon^2 y_2(x) + \dots$$

According to our problem, and for $\epsilon \ll 1$, we can have

$$y(x, \epsilon) \sim y_0(x) + \epsilon y_1(x) + O(\epsilon^2)$$

Then substitute it into our equation $\epsilon y'' = yy' - y$ and expand it: We then perform a substitution into (1):

$$\begin{aligned} \epsilon y_0''(x) + \epsilon^2 y_1''(x) &= y_0(x)y_0'(x) + \epsilon y_1(x)y_0'(x) + \epsilon y_0(x)y_1'(x) + \\ &\quad \epsilon^2 y_1(x)y_1'(x) - y_0(x) + \epsilon y_1(x) + O(\epsilon^2) \end{aligned}$$

Collecting like terms of powers of ϵ :

$$\begin{aligned} \epsilon^2 [y_1''(x) - y_1(x)y_1'(x)] + \epsilon [y_0''(x) - y_1(x)y_0'(x) - y_0(x)y_1'(x) - y_1(x)] \\ = y_0(x)y_0'(x) - y_0(x) + O(\epsilon^2) \end{aligned}$$

$$(y_0 y_0' - y_0) + \epsilon (y_0 y_1' + y_1 y_0' - y_0'' - y_1) + \epsilon^2 (y_1 y_1' - y_1'') = 0$$

We can take the leading order (as ϵ is small) and have

$$y_0 y_0' - y_0 = 0$$

It produces two possible solutions $y_0 = 0$ or $y'_0 = 1$ and implies $y_0 = x + a$ for constant a . According to the paper, “the leading-order solution $y_0(x)$ already fixes the constant by satisfying the boundary condition” and thus a_n for higher order y_n all vanish. In this way, $y_0 = x + a$ can satisfy $\epsilon y'' = yy' - y$ for any ϵ . Based on the left boundary condition and the right boundary condition given, we can have $y_0^L(x) = x + 1$ and $y_0^R(x) = x - 2$. It is also consistent with the observation that there is a boundary layer at either $x = 0$ or $x = 1$.

4.3 Inner Solution

A layer thickness δ , which depends on ϵ , is introduced to help construct “a scaled independent variable that describes positions in the layer” $X \equiv \frac{x-x_0}{\delta}$. (Based on what we have learned in our course, δ is possibly in the form ϵ^α .) Then we can have $Y(X) \equiv y(x) = y(x_0 + \delta X)$. We can observe how y changes in the layer according to X .

$$\frac{dY}{dX} = \delta y', \quad \frac{d^2Y}{dX^2} = \delta y''$$

Then substitute it into our equation $\epsilon y'' = yy' - y$

$$\frac{\epsilon}{\delta^2} \frac{d^2Y}{dX^2} = Y \left(\frac{1}{\delta} \frac{dY}{dX} - 1 \right)$$

That δ should satisfy Y and its derivatives are $O(1)$ as $\epsilon \rightarrow 0$ is mentioned in the paper. It is probably because $Y(X) \sim Y_0(X) + \epsilon Y_1(X) + \dots$. In this way, we can know that ϵ and δ are asymptotically equal, and thus

$$\frac{d^2Y}{dX^2} = Y \frac{dY}{dX} - \epsilon Y$$

$$Y(X) \sim Y_0(X) + \epsilon Y_1(X) + O(\epsilon^2)$$

$$\frac{d^2Y_0}{dX^2} = Y_0 \frac{dY_0}{dX} = \frac{d}{dX} \left(\frac{1}{2} Y_0^2 \right)$$

By integration and separation of variables, we can have

$$Y_0(X) = b \tanh\left(c - \frac{b}{2} X\right)$$

4.4 Zeroth-Order Composite Solutions

4.4.1 Symmetric Solution M

For the interior layer at $x_0 = \frac{1}{2}$, the solution y^M needs to satisfy $y^M(\frac{1}{2}) = Y_0^M(0) = 0$. By the previous Y_0 solution, and Y^M 's symmetry, we can know Y_0^M is an odd function and $c = 0$. Then we need to match the limits of Y^M and the outer solutions. We have $\lim_{X \rightarrow +\infty} Y^M(X) = y_0^R(\frac{1}{2})$ and $\lim_{X \rightarrow -\infty} Y^M(X) = y_0^L(\frac{1}{2})$. By $\tanh(z)$ is odd and $\lim_{z \rightarrow +\infty} \tanh(z) = 1$, we can get $b = \pm \frac{3}{2}$.

$$Y_0^M(X) = -\frac{3}{2} \tanh\left(\frac{3}{4} X\right)$$

By $y_c = y_{outer} + Y_{inner} - y_{match}$, we can have the composite solution

$$y_{c,0}^M = x - \frac{1}{2} - \frac{3}{2} \tanh\left(\frac{3}{4}X\right) + O(\epsilon)$$

4.4.2 Asymmetric Solutions B0 and B1

For B0 which has a boundary layer at $x_0 = 0$, we just need to match $y_0^R(0)$ with $Y_0^{B0}(0)$ and plug in the boundary conditions. We will have

$$Y_0^{B0}(X) = -2 \tanh\left(X - \tanh^{-1}\left(\frac{1}{2}\right)\right)$$

Hence,

$$y_{c,0}^{B0} = x - 2 \tanh\left(\frac{x}{\epsilon} - \tanh^{-1}\left(\frac{1}{2}\right)\right) + O(\epsilon)$$

According to the symmetry between B0 and B1, we can do symmetry transformation $(x, y) \rightarrow (1 - x, -y)$ and get

$$Y_0^{B1}(X) = -2 \tanh\left(X + \tanh^{-1}\left(\frac{1}{2}\right)\right)$$

$$y_{c,0}^{B1} = x - 1 - 2 \tanh\left(\frac{x-1}{\epsilon} + \tanh^{-1}\left(\frac{1}{2}\right)\right) + O(\epsilon)$$

4.5 First-Order Matching

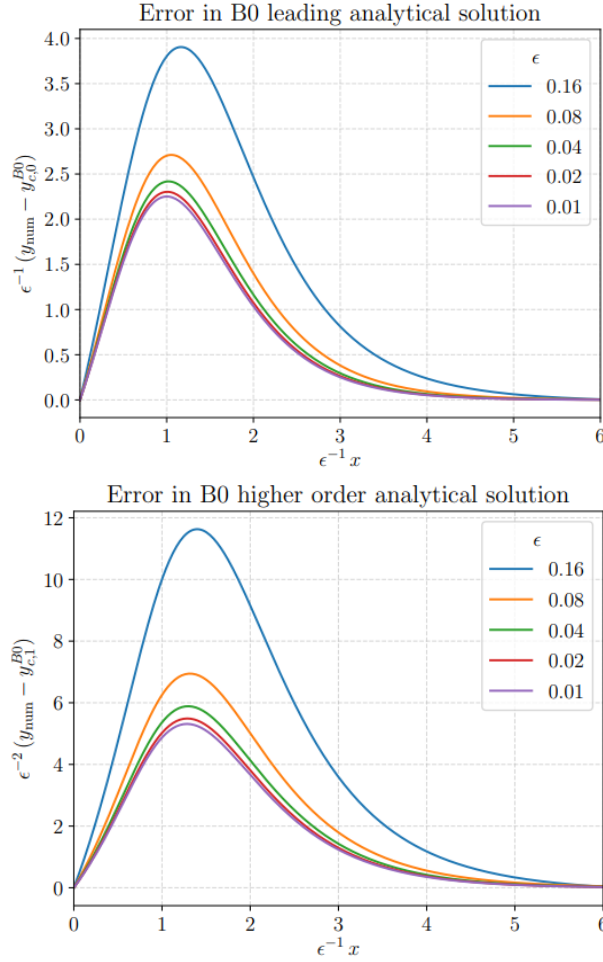
Though we have the zeroth-order composite solutions, they are not precise enough for us to have the study on the initial slope $y'(0)$. It is the motivation for us to do the first-order matching. If we keep Y_1 terms, we can have

$$\frac{d^2 Y_1}{dX^2} - \frac{dY_1}{dX} Y_0 - \frac{dY_0}{dX} Y_1 + Y_0 = 0$$

The authors focus on the symmetric solution M because it has a simpler expression for Y_1 . Using Y_0^M which we already have, we can have the equation

$$\frac{d^2 Y_1^M}{dX^2} + \frac{3}{2} \tanh\left(\frac{3}{4}X\right) \frac{dY_1^M}{dX} - \frac{9}{8} \operatorname{sech}\left(\frac{3}{4}X\right) Y_1^M - \frac{3}{2} \tanh\left(\frac{3}{4}X\right) = 0$$

Due to the overly complex calculation process, here we only elaborate on the idea of the matching process. After organizing the above equation, we can get Y_1^M with unknown constants c_1, c_2 . By $Y_1^M(0) = 0$, we can solve one of them. By matching $\lim_{X \rightarrow +\infty} Y_1^M(X)$ with $y_0^R(\frac{1}{2})$ (by symmetry we can just discuss the right side), we can have the other one. A point worth noting in this process is: if the limit of Y_0^M has already matched y_0^R , we want the constant terms in Y_1^M to vanish because they will lead to $O(\epsilon)$ error if they are non-zero. Then, we can have the composite solution $y_{c,1}^M$. In the similar way we can have $y_{c,1}^{B0}$ and $y_{c,1}^{B1}$. Comparing $y_{c,0}^{B0}$ with $y_{c,1}^{B0}$, we can notice that $y_{c,1}^{B0}$ has significantly smaller errors.



5 Solving Initial Slope

5.1 Reasons to Use a Special Method

To solve the initial slope $y'(0)$, we can utilize the zeroth-order composite solutions and the first-order composite we have solved. For example, for B0, we can have $y_{c,0}^{B0}(0) = 1$ and take derivative to have $y_{c,0}^{B0'}(0) = -\frac{3}{2\epsilon}$. Do the same thing for $y_{c,0}^{B1}$ to reduce error. However, the method we used here for B0 does not work for M and B1. For B0, the method works well because the leading term $-\frac{3}{2\epsilon}$ will be unchanged and dominant while we going to the next order. For M, we will find that going to the next order will lead to changes to the dominant term, which makes the solutions not match. This issue implies that we should use a method that can utilize more information.

The method used in the paper is to employ a conserved quantity so that we can utilize the information about the structure of solutions in the phase plane. It can transfer information from the layer to a distinct point all the way.

5.2 Solving

From what we have learned in the phase portrait, we can have

$$\frac{dz}{dy} = \frac{z'(t)}{y'(t)} = \frac{y(z-1)}{\epsilon z}$$

It yields

$$2\epsilon[z + \log(1-z)] - y^2 = C^2$$

for a constant C . It shows that as $y(t)$ and $z(t)$ flow along a trajectory, $2\epsilon[z + \log(1-z)] - y^2$ will be unchanged, and thus it's called a conserved quantity. Then, suppose we have a point (x_1, y_1) , and we can solve z_0 at another point on the same curve through

$$2\epsilon[z_0 + \log(1-z_0)] - y_0^2 = 2\epsilon[z_1 + \log(1-z_1)] - y_1^2$$

Suppose $y_0 = 1$, we can solve z_0 with Lambert W function

$$1 - z_0 = -W_0(-[1 - z_1] \exp(\frac{1 - y_1^2}{2\epsilon} - (1 - z_1)))$$

For B1, by its symmetry with B0, we know that B1's final slope is B0's initial slope which we can solve. So by the procedure we mentioned, we can have $y_{B1}(1) = y_{c,1}^{B_0}(0) = -1$ and $z_{B1}(1) = y_{c,1}^{B_0'}(0) = -\frac{3}{2\epsilon} + 1 + \log(16) + O(\epsilon)$. Then we can solve for $y'_{B1}(0)$. We can do the same thing for M but choose z_1 from the the first-order composite solution at $x = \frac{1}{2}$.

6 The Bifurcation Issue

6.1 What is the pitchfork bifurcation?

One of the most important aspects of this problem to assess is what happens to each of these solutions slopes as ϵ leaves 0. Graphing the solutions we can clearly see that for sufficiently small ϵ there is a natural ordering: $y'_{B_0}(0) < y'_M(0) < y'_{B_1}(0)$. However, examining the solutions we can see that as ϵ increases, there is some sort of merging that happens. Looking at the figure below, we can see this pitchfork bifurcation occurring both in a scaled sense to showcase the merge as well as what the solutions look like in a more global sense.

It should be noted that these graphs are the contour lines $(1 + y(1))$ in the $(\epsilon, y'(0))$ plane. Now we examine where $y(1) + 1 = 0$ which occurs between the red and blues in figure 4. Below the critical value $\epsilon < \epsilon_e$, there are three solutions that bifurcate at $\epsilon = \epsilon_e$ and $y'(0) = 0$. However, for large values of

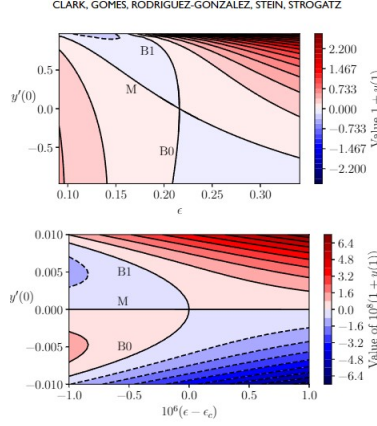


Figure 5: Contour Lines and the Pitchfork Bifurcation

ϵ there is still only one solution. So how do we explain the bifurcation that occurs? Well we can understand that the relationship between the initial and final slopes of the B0 and B1 solutions are paired under symmetry, as described previously. We can take the two points (y_0, z_0) and (y_1, z_1) to be endpoints on the symmetry trajectory we showcased previously. Taking $y_0 = 1$ and $y_1 = -1$ we see that:

$$z_0 + \log(1 - z_0) = z_1 + \log(1 - z_1)$$

Now we can define a function:

$$f(z) = z + \log(1 - z)$$

But as the authors observe, the symmetric solutions tell us nothing. Therefore, we look at the asymmetric solutions on the endpoints. Looking at the shape of the graph of f we can see there is only one local extremum at $z = 0, f(0) = 0$. The second derivative: $f''(z) = -(1 - z)^{-2}$ is strictly negative so we can conclude that $z = 0$ is a global maximum. For $z < 0$ f is monotonically increasing and in $0 < z < 1$ $f(z)$ is monotone decreasing. We can clearly see that because they are asymmetric and monotonic increasing and decreasing in these intervals, we know that the two slopes must live in $z_0 < 0$ and $0 < z_1 < 1$. Looking at the graph we can also see that the global maximums which are cut at two places ultimately converge equal at $z = 0$. Ultimately this tells us that $y'(0) = z_{B_0}$ and $y'(1) = z_{B_1}$ is the solution to B0 and B1 is the opposite. Clearly as ϵ increases they slopes approach each other until they meet at $z_{B_0} = z_{B_1} = 0$ at $\epsilon = \epsilon_e$

The M solution is equally interesting. We clearly want to know after looking at the graphs whether or not this solution will ever merge with these other two solutions. The answer is clearly no and this is because they must all merge

simultaneously and it can't just be a mix and match. This is shown by the small error function at the bifurcation point which the authors posit is cubic: $y(1) + 1 \approx A(y'(0))^3 + B(\epsilon - \epsilon_e)y'(0)$. Ultimately, any expectation or viewpoint will ultimately confirm the pitchfork.

6.2 Calculating it's value

But how do we calculate this value? The bifurcation happens when we get the solution $y'(0) = 0$ so we can use that to find ϵ_e . This must also satisfy the boundary condition. We can see using the trajectories and checking that it satisfies our log equation to get the critical trajectory $C = 1$. We then get a condition on ϵ_e that $(y, z) = (0, z_c)$ with $C = 1$ giving us:

$$2\epsilon_e[z_e + \log(1 - z_c)] = -1$$

We then see that we must also satisfy the time condition of the critical trajectory: $y = 1$ to $y = -1$ is $t = 1$. However this trajectory is also self symmetric which tells us that the time to go halfway is $t = \frac{1}{2}$ which is also the point $(y, z) = (0, z_c)$. Neat! Now, we have the integral:

$$\frac{1}{2} = \int_0^{\frac{1}{2}} dt = \int_0^{z_c} \frac{\epsilon_e dz}{y(z-1)}$$

Taking the positive root of the trajectory when $C = 1$ for $2\epsilon_e[z + \log(1 - z)] = y^2 - 1$ we get:

$$y = \sqrt{1 + 2\epsilon_e[z + \log(1 - z)]}$$

Which gives us a travel time condition:

$$\frac{1}{2} = \int_{z_c}^0 \frac{\epsilon_e dz}{(1 - z)\sqrt{1 + 2\epsilon_e[z + \log(1 - z)]}}$$

Which after too much algebra we get:

$$0 = g(z_c) = z_c + \log(1 - z_c) + \int_{z_c}^0 \frac{dz}{(1 - z)\sqrt{1 - \frac{z + \log(1 - z)}{z_c + \log(1 - z_c)}}}$$

where $z_c \approx -3.905$ and we get $\epsilon_e \approx 2.16$

7 Conclusion

In summary, these findings can tell us a lot not just about examining error functions as important catalysts for interesting math problems, but also how certain questions, when revisited, can glean extremely interesting multidimensional solutions not originally posited. This question in particular, though specific, can

help us look at problems in engineering and physics that give us a better or more specific understanding of a particular problem. Moreover, this problem in particular shows the value in determining solutions to singularly perturbed differential equations as problems that directly impact and motivate scientific inquiry.